

4.1: Matrix Algebra

- 7) Let A have the block form

$$A = \begin{bmatrix} B & C \\ 0 & I \end{bmatrix}$$

in which the blocks are $n \times n$. Prove that if $B - I$ is nonsingular, then, for $k \geq 1$,

$$A^k = \begin{bmatrix} B^k & (B^k - I)(B - I)^{-1}C \\ 0 & I \end{bmatrix}$$

- 12) Let A be an $n \times n$ invertible matrix, and let u and v be two vectors in $\mathbb{R}^n$. Find the necessary and sufficient conditions on u and v in order that the matrix

$$\begin{bmatrix} A & u \\ v^T & 0 \end{bmatrix}$$

be invertible, and give a formula for the inverse when it exists.

- 17) A square matrix A is said to be skew-symmetric if $A^T = -A$. Prove that if A is skew-symmetric, then $x^T Ax = 0$ for all x .
- 18) (Continuation) Prove that the diagonal elements of a skew-symmetric matrix are 0. Also, prove that the determinant is 0 when the matrix is of odd order.
- 19) (Continuation) let A be any square matrix, and define $A_0 = \frac{1}{2}(A + A^T)$ and $A_1 = \frac{1}{2}(A - A^T)$. Prove that A_0 is symmetric, that A_1 is skew-symmetric, that $A = A_0 + A_1$ and that for all x , $x^T Ax = x^T A_0 x$. This explains why, in discussing quadratic forms, we can confine our attention to **symmetric matrices**.

4.2: LU and Cholesky Factorizations

- 1) Proof these facts, needed in the proof of Theorem 2.
 - (a) If U is upper triangular and invertible, then U^{-1} is upper triangular.
 - (b) The inverse of a unit lower triangular matrix is unit lower triangular.
 - (c) The product of two upper (lower) triangular matrices is upper (lower) triangular.
- 5) Prove that an upper or lower triangular matrix is nonsingular if and only if its diagonal elements are all different from 0.